



ASSIGNMENT NUMBER 1 PART 1

MOSIMA MABEL MALEFO



1. Introduction

The purpose of this analysis is to use the lifestyle and demographics features from the insurance dataset to predict the cost of insurance. to guarantee accuracy and interpretability of data, linear aggression approach was used along with exploratory data analysis, feature selection, training the model and evaluating the model.

1.2 Libraries that I have used to analyses this dataset

I have imported pandas to work with structured data in form of tables, NumPy to work with numerical data, seaborn to help with visualization and matplotlib to present the results clearly in the report using different types of graphs.

1.3 Loading dataset

This code `df = pd.read_csv("insurance.csv")` was used to load dataset into the jupyter notebook environment for preparation analysis and modeling the data.

1.4 Description of dataset

age of primary insurance owner.

Sex the gender of the insurance owner whether is female or male.

Bmi – understanding the weight of the owner which is relative to height.

Children- the number dependent covered by the health insurance.

Smoker- checking the health insurance owner is smoking.

Region- the health insurance owner's residential address.

Charges- medical cost of the insurance.

According to wikipedia Contributors (2019) Exploratory Data Analysis- is a technique of data set analysis that uses statistical charts and other data visualization techniques to sum up the main features of the data.

1.5 Inspecting the dataset

Head of the dataset

Info of the dataset

Description of dataset

shape of the dataset

```
print(df.head())
print(df.info())
print(df.describe())
```

```

   age  sex  bmi  children  smoker  region  charges
0   19  female  27.900      0      yes  southwest  16884.92400
1   18  male  33.770      1      no  southeast  1725.55230
2   28  male  33.000      3      no  southeast  4449.46200
3   33  male  22.705      0      no  northwest  21984.47061
4   32  male  28.880      0      no  northwest  3866.85520
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   age          1338 non-null   int64
1   sex          1338 non-null   object
2   bmi          1338 non-null   float64
3   children     1338 non-null   int64
4   smoker       1338 non-null   object
5   region       1338 non-null   object
6   charges      1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
None

```

	age	bmi	children	charges
count	1338.000000	1338.000000	1338.000000	1338.000000
mean	39.207025	30.663397	1.094918	13270.422265
std	14.049960	6.098187	1.205493	12110.011237
min	18.000000	15.960000	0.000000	1121.873900
25%	27.000000	26.296250	0.000000	4740.287150
50%	39.000000	30.400000	1.000000	9382.033000
75%	51.000000	34.693750	2.000000	16639.912515
max	64.000000	53.130000	5.000000	63770.428010

Head function was used to display first five rows of the data to check the structure of the dataset, see the columns names and values and to understand how the data is organized.

Info function was used to see the summary of dataset. Checking data types of the dataset to determine which one is numerical and category and check the missing values.

Describe(s) function was used to produce the summary of statistics for numerical value in the dataset this include quartiles, standard deviation, mean and minimum and maximum values in the dataset.

2. Shape of the dataset before and after converting non-numerical data to numerical data.

```
df.shape
```

```
(1338, 7)
```

The shape of the dataset show that we have 1338 rows and 7 columns

Numerical features

```
[features for features in df.dtypes[df.dtypes != 'object'].index]
```

```
['age', 'bmi', 'children', 'charges']
```

I have used this code to check my numerical dataset, and it shows that I have four numerical datasets which are age, bmi, children and charges.

2.1 Categorical features

```
cat_features = df.dtypes[df.dtypes == 'object'].index  
cat_features
```

```
Index(['sex', 'smoker', 'region'], dtype='object')
```

I have used this code to check categorical dataset, and it shows that I have three categorical datasets which are sex, smoker, and region.

df.shape the original dataset was (1338, 7) and I have changed non numerical data to numerical data on the region I had to add 4 columns to accommodate the data that I have and I had to drop the original region so that the dataset won't see it as duplicate now the shape of the dataset is (1338, 12).

2.2 Checking for Missing Values

```
[5]: print(df.isnull().sum())
```

```
age      0  
sex      0  
bmi      0  
children 0  
smoker   0  
region   0  
charges  0  
dtype: int64
```

The dataset is clean and it does not have any missing values.

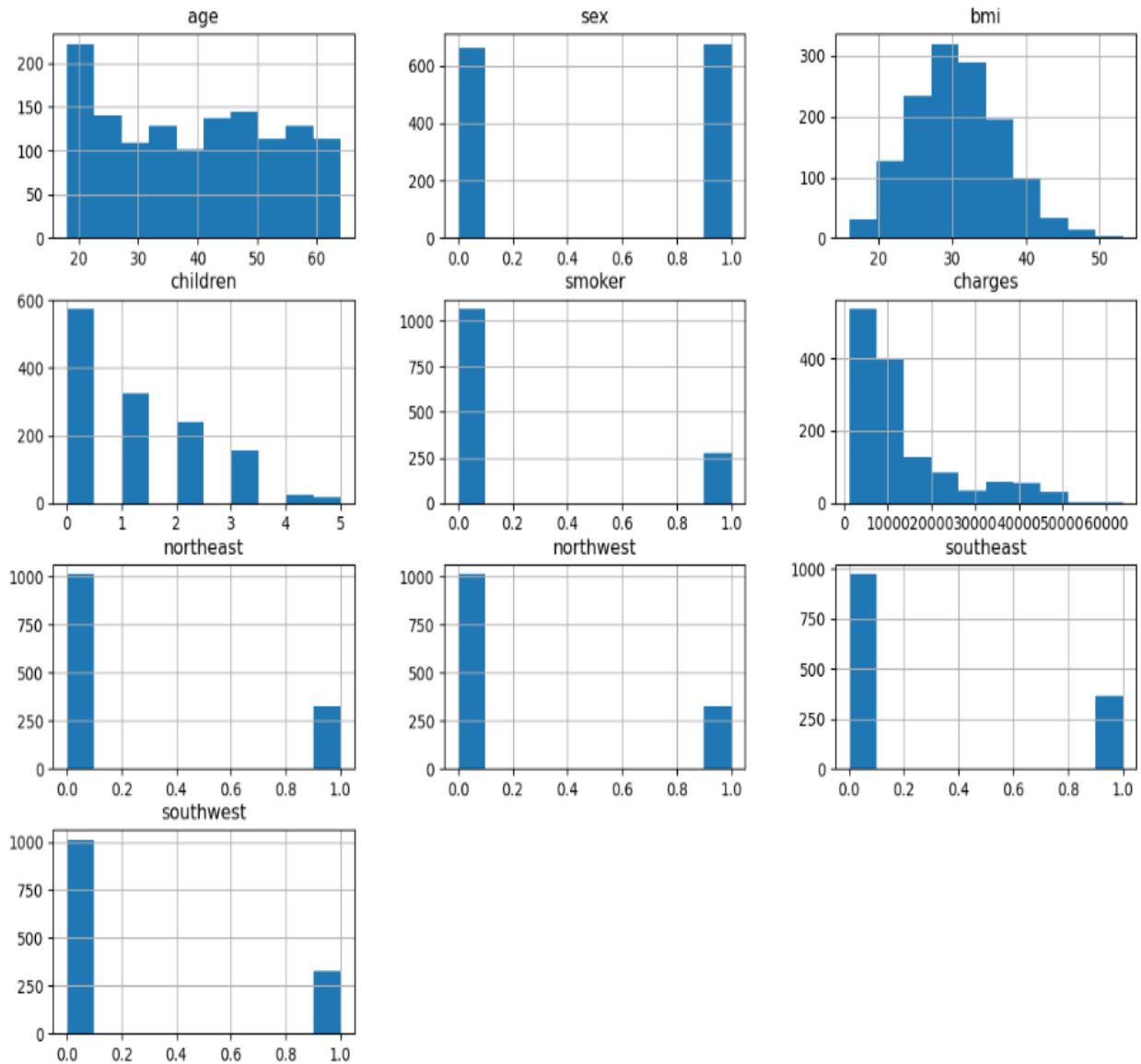
3. Converting columns from string (sex, smoker, region) to numerical values

	age	sex	bmi	children	smoker	charges	northeast	northwest	southeast	southwest
0	19	0	27.900	0	1	16884.92400	0	0	0	1
1	18	1	33.770	1	0	1725.55230	0	0	1	0
2	28	1	33.000	3	0	4449.46200	0	0	1	0
3	33	1	22.705	0	0	21984.47061	0	1	0	0
4	32	1	28.880	0	0	3866.85520	0	1	0	0
...	...	...	...	...	...	...	...	...	...	...
1333	50	1	30.970	3	0	10600.54830	0	1	0	0
1334	18	0	31.920	0	0	2205.98080	1	0	0	0
1335	18	0	36.850	0	0	1629.83350	0	0	1	0
1336	21	0	25.800	0	0	2007.94500	0	0	0	1
1337	61	0	29.070	0	1	29141.36030	0	1	0	0

1338 rows × 10 columns

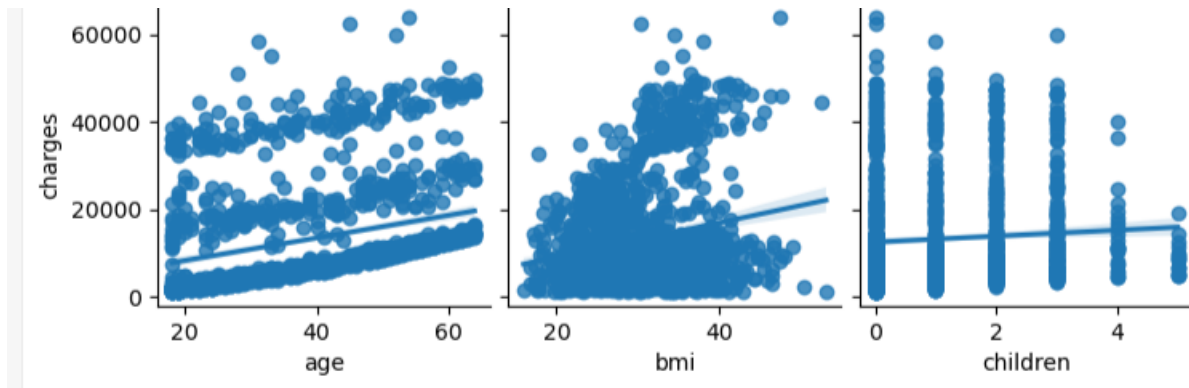
Linear regression model understands numerical data. So, I have converted categorical data into numerical data to ensure that the model can process data correctly. Map() function was used to convert categorical values to numbers in sex: female = 0, male =1, smoker: No =0, yes = 1. Pd.get_dummies() function was used to separate the columns of the category region into 4 columns with values 0/1 and the region was dropped to remove the column after encoding to prevent the model from thinking that there is a relationship between the regions.

4. Visualizations of all the variables

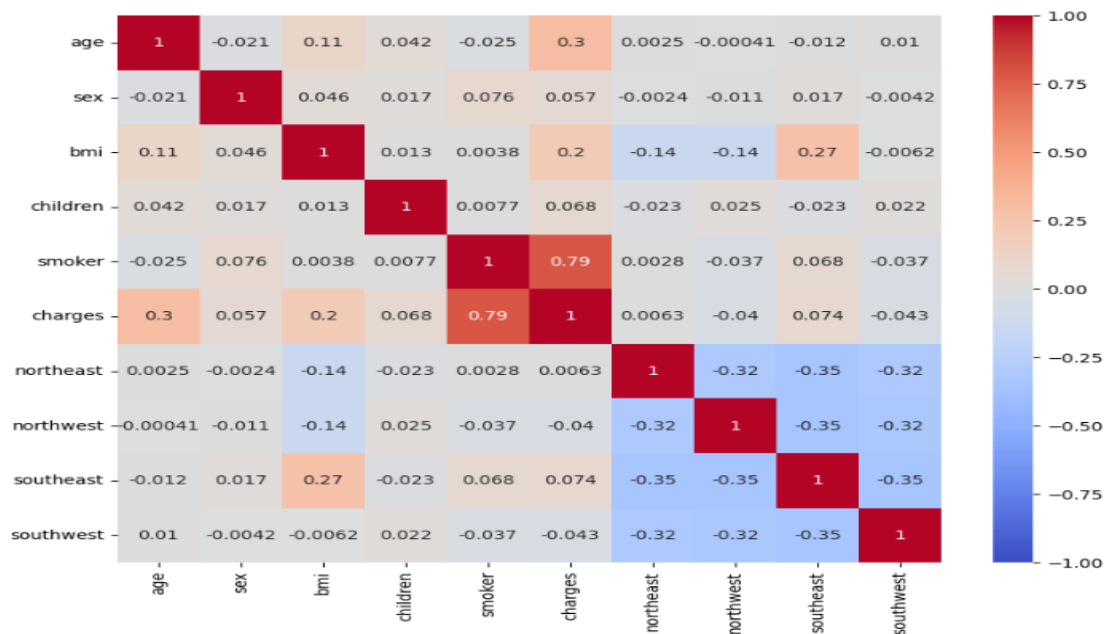


I have visualized all the variables to help understand the relationship between them before creating the linear regression model. This process helps with feature selection and preprocessing to ensure that the model is trained on data that is clear, well understood and clean because all data is converted into numerical data. By analyzing these graphs, one can check which variables may have significant impact on the charges and which one require modification. This step is necessary for conducting exploratory data analysis in a way that is productive, since it allows for inspection of distribution and correlations in advance of linear regression modeling.

5. Relationship between features and target

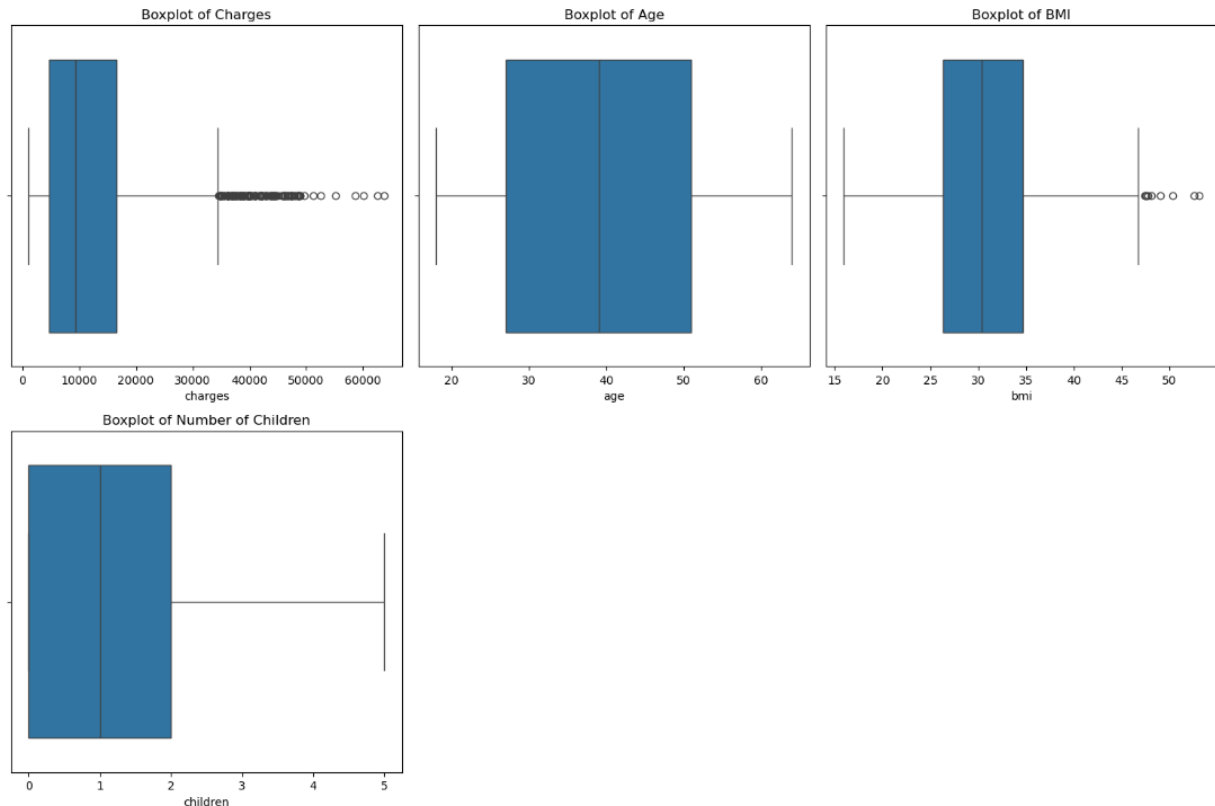


According to wikipedia Contributors (2019) Scatterplots is a kind of graph or mathematical plot that uses Cartesian coordinates to represent data values for usually two variables. Scatter plots were used to check the relationship between age, bmi, children and charges. Age against the charge shows that older people pay more on insurance than children because older people are prone to health problems which is a risk factor. Bmi is also associated with higher insurance charges because it is frequently linked to being overweight which raises the risk of health issues and increase insurance costs. Age scatter plot.



A correlation heatmap was created to analyze the connections between numerical variables in the dataset. The correlation shows that age, smoker and Bmi are highly associated with more charges because they high risk factors. Children have mild correlations. Regions have small impact on charges on insurance.

5.1 Check for Outliers



Outliers are sensitive to linear regression, so I have used boxplots. `plt.tight_layout()` function is used to give the distance between the subplots to automatically adjust to overlapping titles and axes. `plt.show()` function shows the plots.

Boxplot of charges shows extremely high outliers that point to very high insurance payment for certain people. The distribution is plainly skewed. Age shows unusual on old people and young people are outliers. BMI values are outliers and may have an impact on the insurance risk.

Boxplots offer a brief overview of the distribution and outliers for important numerical characteristics. Data preprocessing choices, like log transformations or outlier treatment, which increase the precision of linear regression modeling, are guided by boxplot observations.

5.2 Splitting and testing

Linear regression learns patterns from the data. I want to see how accurately it predicts new and unseen data. For this reason, I have divided the dataset.

```
y_train
560      9193.83850
1285     8534.67180
1142     27117.99378
969      8596.82780
486      12475.35130
...
1095     4561.18850
1130     8582.30230
1294     11931.12525
860      46113.51100
1126     10214.63600
Name: charges, Length: 1070, dtype: float64
```

The training set I have used 80% of the dataset to train the model then link them between features (x) and the target (y) is learned by the model.

5.3 Testing the dataset

20% remaining portion is used to assess how the model performed on remaining. This enabled me to access whether the model it overfits.

5.4 Importing the model

I have used sklearn import models such linear regression, SVR, Random Forest regression and Gradient Boosting Regressor.

5.5 Training the model

I have imported NumPy to train_x and train_y. Gradient Boosting Regressor is best performing parameters.

```
RandomForestRegressor
Parameters
```

I have used this code to predict the charges

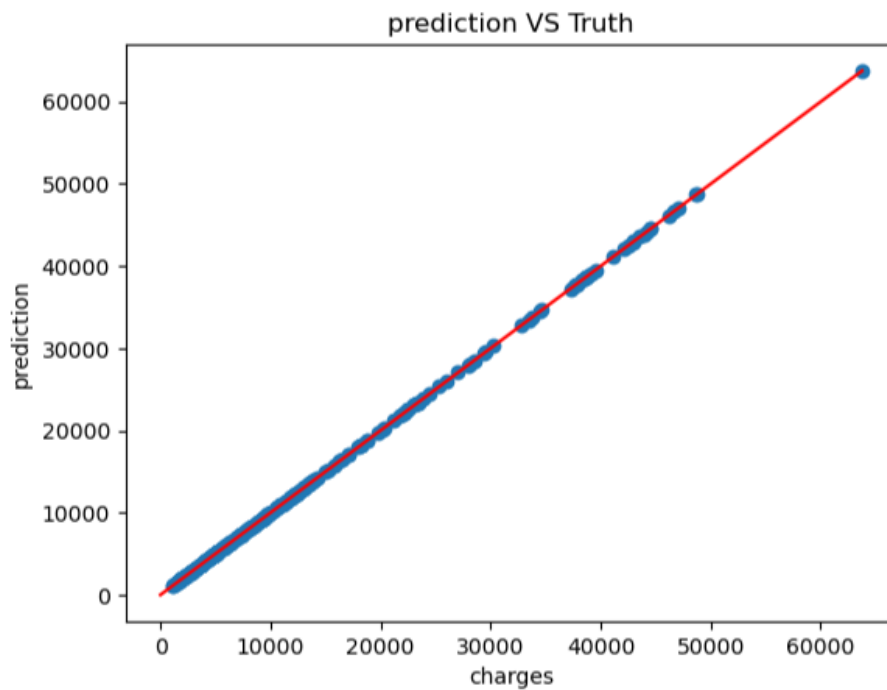
```
LR.predict(x_test)
svm.predict(x_test)
RFR.predict(x_test)
GBR.predict(x_test)
```

I have used this to compare the tables in the dataset

```
df1 = pd.DataFrame({  
'Actual': y_test,  
'LR': y_pred1,  
'svm': y_pred2,  
'RFR': y_pred3,  
'GBR': y_pred1,
```

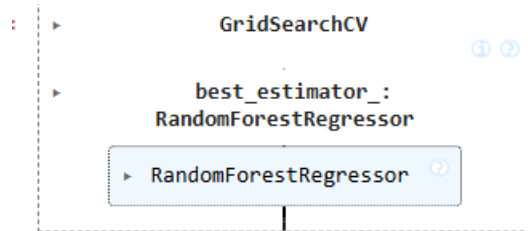
df1 visualize the model predictions.

6. Visualizing Model Predictions vs Actual Charges



I have used the scatter plot to check predicted values vs the actual values. The red line shows perfect predictions. Everything on the line means my prediction is right. This algorithm is used to test all possible combinations and find the best model configuration.

7. Hyperparameter



I have used grid search for my hyperparameter is this step is crucial for improving model performance and generalization.

8. References

1. Wikipedia Contributors (2019). Exploratory data analysis. [online] Wikipedia. Available at: https://en.wikipedia.org/wiki/Exploratory_data_analysis.